

Scalability & Future Plan

Date	2 July 2026
Team ID	XXXXXX
Project Name	RISING WATERS
Maximum Marks	1 Mark

Scalability & Future Plan:

The RISING WATERS project is designed to be scalable and can be enhanced to support larger datasets, more users, and real-time flood monitoring. Future improvements include IoT sensor integration, satellite data analysis, mobile alerts, and advanced AI models for more accurate flood prediction

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited real-time data sources	Prediction accuracy may decrease	High
2	No mobile notification system	Users may miss emergency alerts	Medium
3	Limited geographical coverage	Supports only selected region	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users	Deploy cloud-based infrastructure for more users
2	Data Storage	Stores historical data only	Use cloud databases for large-scale storage
3	Performance	Basic AI prediction model	Optimize AI model for faster prediction
4	Security	Basic authentication	Implement secure login and data encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 1	Real-time rainfall and river data integration	3 Months	More accurate prediction
Phase 2	Mobile App with instant flood alerts	6 Months	Faster emergency response
Phase 3	Satellite & IOT sensor integration	9 Months	Improved monitoring and accuracy
Phase 4	Advanced AI-based flood forecasting	12 Months	Higher prediction accuracy and wider coverage